



Adoption of Artificial Intelligence in Facilitating the Professional Development of Lecturers



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Abstract: Artificial intelligence (AI), particularly generative AI, is rapidly transforming teaching methodology in universities as well as their assessments, research, and academic work, yet there is only limited evidence about how lecturers in sub-Saharan African universities adopt these tools and what professional development they require. This qualitative study explored AI adoption, perceived benefits and risks, and needs for professional development among lecturers and administrators in the Faculty of Education at one private university in Uganda. A convenience sample of 28 participants took part in individual semi-structured interviews during 14-day of campus visit in January, February, and March 2026. The interview guide was pilot tested; while interviews lasted for 38–67 minutes each, meaning saturation was judged to have been reached by the 25th interview, with three additional interviews required to assess the adequacy of the developing coding framework. Reflexive thematic analysis generated 6 themes: pragmatic but uneven adoption; AI as partner for efficiency and creativity; uncertainty about accuracy, authorship, and academic integrity; infrastructural and institutional constraints; demand for practice-based and discipline-relevant professional development; and the need for governance, communities of practice, and protected learning time. Lecturers more often framed AI through teaching, assessment, and workload concerns, whereas administrators more often foregrounded policy consistency, governance, and institutional support. Participants reported using AI frequently for lesson planning, summarizing, language editing, idea generation, assessment preparation, research support, and routine administration. However, adoption was constrained by unreliable connectivity, subscription costs, uneven AI literacy, limited policy guidance, privacy concerns, and fear of students' overreliance. The study proposed a contextualized professional-development model combining foundational AI literacy, pedagogical design, research integrity, data protection, assessment redesign, peer mentoring, and continuing technical support. The findings provide a situated account of responsible AI adoption in an East African Faculty of Education and should not be interpreted as representative of Ugandan higher education as a whole.

Keywords: Artificial intelligence; Generative artificial intelligence; Professional development; Higher education; Lecturers

1. Introduction

Artificial intelligence (AI) has developed from a specialized technical domain to an everyday component of academic work. Generative AI systems could produce text, images, code, explanations, feedback, and other outputs in response to natural-language prompts. In higher education, these capabilities are being used for teaching preparation, student support, assessment design, research, administration, and scholarly communication. Recent reviews have pointed out that AI applications in higher education now include assessment and evaluation, prediction, intelligent tutoring, AI assistants, and student-learning management, while generative AI has intensified debates about authorship, reliability, fairness, and the purposes of university assessment (Crompton & Burke, 2023; Kasneci et al., 2023; UNESCO, 2023).

The speed of adoption has created a mismatch between access to AI tools and the availability of professional learning, institutional guidance, and governance. Faculty members may experiment independently, but meaningful educational use requires more than knowing how to enter a prompt. It requires AI literacy, pedagogical judgement, awareness of bias and hallucination, knowledge of privacy and copyright, and the capacity to redesign learning activities and assessments. In this study, AI literacy refers to the knowledge, practical skills, critical judgement, and ethical awareness required to understand the capabilities and limitations of AI systems, formulate and refine prompts where appropriate, verify outputs, accept risks relating to bias, privacy, and authorship, and make pedagogically appropriate decisions about AI use. The United Nations Educational, Scientific and Cultural Organization (UNESCO)'s guidance and competency framework place human agency, ethics, pedagogy, and professional learning at the core of teacher's AI competence (UNESCO, 2023; UNESCO, 2024). Similarly, faculty-focused research indicated that use, confidence, and requirements for professional development varied substantially across instructors, hence suggesting that uniform training was unlikely to be sufficient (Mah & Groß, 2024; Uzorka et al., 2023).

These issues are especially important in Faculties of Education in tertiary institutes. Lecturers in such faculties influence both university students and prospective teachers introducing AI-related practices into schools. Their professional development therefore has a multiplier effect. Meanwhile, universities in low- and middle-income settings often face connectivity constraints, high data costs, limited access to licensed tools, and uneven technical support. These contextual conditions could be conducive to a form of adoption that is individually driven and highly pragmatic but institutionally fragile. Emerging work in African higher education spotlighted enthusiasm about AI, alongside concerns about infrastructure, policy, academic integrity, and staff capability; however, context-specific qualitative evidence from Uganda remained sparse. The challenge to teachers' professional development is consequently not only technical but pedagogical, ethical, epistemic, and organizational.

The purpose of this study is to explore how lecturers and administrators in the Faculty of Education at a Ugandan university understand and apply AI, the barriers and risks they encounter, and the forms of professional development and institutional support they consider necessary for responsible adoption. Under the guidance of the above objective, this study addressed the following three research questions:

RQ1. How are lecturers and administrators adopting AI in teaching, research, assessment, and academic administration?

RQ2. What benefits, risks, and barriers do participants associate with AI adoption?

RQ3. What approaches to professional development and institutional conditions do participants consider necessary for responsible and sustainable AI adoption?

2. Literature Review

2.1 Artificial Intelligence in Higher Education

AI in higher education encompasses a broad set of technologies, including learning analytics, recommendation systems, automated assessment, intelligent tutoring systems, predictive models, conversational agents, and generative AI. A systematic review by Crompton & Burke (2023) discovered expanding utilization of AI across teaching and learning functions but also noted that evidence was unevenly distributed across disciplines and regions. A meta-review similarly called for stronger ethics, collaboration, and methodological rigour, underscoring the need to move beyond technologically deterministic accounts of educational innovation (Bond et al., 2024).

Generative AI has expanded the accessibility and visibility of AI because users could interact with the systems through ordinary language. Potential educational benefits include rapid feedback, brainstorming, translation, personalization, simulation, accessibility support, and assistance with routine academic tasks. Nevertheless, large language models may generate plausible but incorrect information, reproduce social biases, fabricate sources, and encourage uncritical dependence. Their outputs therefore require verification, disciplinary knowledge, and explicit pedagogical framing (Kasneci et al., 2023; Lo, 2023; UNESCO, 2023).

For lecturers, AI could support course planning, creation of examples, generation of questions, differentiated explanations, rubric development, and communication. In research, it could assist with the development of ideas, coding support, editing, literature discovery, and data-analysis workflows. Yet, the boundary between assistance and inappropriate substitution is often unclear. Faculty members should make decisions about disclosure, authorship, confidential data, intellectual property, and acceptable student use. These decisions are difficult to make when institutional policies are absent, rapidly changing, or framed only as prohibition.

2.2 Faculty Adoption, Self-efficacy, and Professional Development

Technology adoption is influenced by perceived usefulness, ease of use, self-efficacy, social norms, access to resources, and organizational support. In the case of AI, these factors interact with ethical and epistemic concerns. Mah & Groß (2024) identified distinct faculty profiles in relation to AI use and self-efficacy; the scholars

demonstrated that professional-development needs differed among confident users, cautious experimenters, and less-engaged staff. This finding supported differentiation rather than one-size-fits-all professional development.

Professional development for employing AI should integrate technological knowledge with pedagogical and disciplinary knowledge. Training restricted to demonstrations of popular tools risks becoming obsolete as products change and may encourage superficial use. More durable learning outcomes include understanding what AI systems could and could not do, designing prompts and verification routines, aligning AI use with learning outcomes, redesigning assessment, protecting personal and institutional data, and evaluating bias and accessibility. UNESCO's AI Competency Framework for Teachers identifies 5 related dimensions: a human-centred mindset, ethics of AI, AI foundations and applications, AI pedagogy, and AI for professional learning (UNESCO, 2024). These dimensions inform the use of AI literacy in this study, but AI literacy is treated as an integrated professional capability rather than as prompt-writing skill alone. It includes foundational understanding, critical verification, ethical and data-aware judgement, pedagogical application, and continuing professional learning.

Adult professional learning is more effective when it is relevant to immediate work, active, collaborative, sustained, and supported by opportunities for reflection and practice (Olaniyan & Uzorka, 2024). For university lecturers, this refers to short demonstrations followed by guided experimentation, discipline-specific examples, peer mentoring, and continuing consultation. Faculty should also be involved in the design of institutional AI policies because responsible adoption depends on academic values, local assessment cultures, and shared governance rather than technical compliance alone.

2.3 Ethics, Academic Integrity, and Governance

Academic integrity has dominated early institutional responses to generative AI. Concerns include undisclosed machine-generated work, fabricated references, contract-cheating-like substitution, and the predicament of establishing authorship. However, a narrow focus on detection could be counterproductive because AI detectors are unreliable and could generate false positives. Responsible governance instead requires transparent expectations, assessment redesign, process evidence, oral or performative components where appropriate, and explicit instruction in ethical AI use.

Ethical concerns extend beyond cheating. AI services may collect prompts and uploaded materials, creating risks where lecturers enter student records, unpublished research, examination questions, or identifiable personal information. Bias could also affect examples, feedback, language, and representation. UNESCO (2023) therefore recommended human-centred governance, data protection, age-appropriate use, validation, and attention to inclusion. Systematic reviews of fairness, accountability, transparency, and ethics in AI and higher education similarly emphasize the need for explainability, human oversight, and equitable access (Akgun & Greenhow, 2022; Bond et al., 2024; Ogunleye et al., 2024).

Institutional guidance should be appropriately clear to support action but flexible enough to accommodate disciplinary differences. Policies should distinguish permitted, restricted, and prohibited uses; require disclosure where AI materially contributes; address data protection and procurement; and define accountability for high-stakes decisions. Professional development is the mechanism through which such policy becomes practical.

2.4 Artificial Intelligence Adoption in African and Resource-Constrained Higher Education

The opportunities associated with AI are global, but conditions for adoption are unequal. In a large number of African universities, mobile connectivity has widened digital participation, yet bandwidth, device quality, electricity reliability, and subscription costs remain consequential. AI tools that operate through cloud services could intensify these inequalities because advanced features are often paywalled and data intensive. English-dominant training data may limit the relevance of outputs for local languages, curricula, and contexts. These conditions provide infrastructure, affordability, and institutional support of the adoption process rather than merely background constraints.

Uganda's policy environment provides a more specific context for these concerns. The Ministry of Education and Sports' Education Digital Agenda Strategy 2021–2025 set out a national rationale and action plan for integrating information and communication technology (ICT) into teaching, learning, assessment, and administration (Ministry of Education & Sports, 2021). More recent national AI-readiness work has similarly highlighted governance, data protection, digital infrastructure, skills development, innovation, and inclusion as central issues in Uganda's emerging AI ecosystem (Ministry of ICT and National Guidance & International Telecommunication Union, 2025). These priorities are directly relevant to universities because staff adoption takes place within wider national conditions of connectivity, digital capability, regulation, and institutional readiness.

Empirical work from Uganda is also beginning to show how local adoption differs from accounts based mainly on high-income settings. Namatovu & Kyambade (2025), studying ChatGPT adoption among university students in Uganda, discovered that performance expectancy, habit, social influence, and facilitating conditions were important to adoption and noted the continuing relevance of digital-literacy and technological-access constraints.

Although their study focused on students rather than lecturers, it demonstrates that AI uptake in Ugandan higher education is shaped by both perceived usefulness and the resources and social conditions surrounding use. This reinforces the need for lecturer-focused qualitative evidence that examines not only whether AI is used, but how institutional, pedagogical, and professional-development conditions shape responsible use.

Faculty development in such settings should therefore be designed around available infrastructure and institutional priorities. Low-bandwidth options, shared access, locally relevant cases, and careful tool selection are essential. Moreover, professional development should not frame lecturers as deficient users who should simply catch up. Their concerns about workload, epistemic quality, cultural relevance, student learning, privacy, and institutional legitimacy are legitimate design inputs. The current study addressed this gap by examining AI adoption as a situated professional-practice issue in one Faculty of Education in Uganda.

2.5 Positioning and Contributions of the Present Study

Recent international literature has established that faculty AI use is growing, that self-efficacy and professional-development needs vary, and that ethics and governance are central. Uganda's education policy has increasingly accentuated digital integration, and emerging Ugandan research has begun to examine AI adoption among university students (Ministry of Education & Sports, 2021; Namatovu & Kyambade, 2025). Yet three little studied areas remain. First, there is still limited qualitative evidence on how lecturers and academic administrators in Ugandan and East African Faculties of Education are incorporating generative AI into everyday professional practice. Second, a large number of studies focused on attitudes or tool use without connecting adoption to the practical design of professional development. Third, resource and institutional conditions are often treated as background variables rather than constitutive features of adoption. This study contributes a contextualized account linking actual and anticipated uses, perceived risks, structural barriers, role-based differences, and preferred professional-learning formats. The summary of previous work related to AI adoption and the professional development of lecturers is presented in Table 1.

Table 1. Summary of previous works related to artificial intelligence (AI) adoption and professional development (PD) of lecturers

Study	Context and Design	Key Relevance
Crompton & Burke (2023)	Systematic review of AI in higher education	Maps major AI uses and highlights gaps in disciplines and contexts.
Kasneji et al. (2023)	Interdisciplinary review of large language models in education	Identifies opportunities alongside hallucination, bias, and overreliance risks.
UNESCO (2023)	Global guidance	Provides human-centred, ethical, policy, and capacity-building principles.
Mah & Groß (2024)	Faculty survey; $N = 122$	Delineates heterogeneous faculty profiles, self-efficacy, use, and PD needs.
UNESCO (2024)	AI competency framework for teachers	Defines competency dimensions for teacher training and professional learning.
Ministry of Education & Sports (2021)	Uganda; National Education Digital Agenda Strategy	Provides national policy context for information and communication technology (ICT) integration in teaching, learning, assessment, and administration.
Namatovu & Kyambade (2025)	Uganda; mixed-method study of university students; $N = 473$	Demonstrates that AI adoption in Ugandan higher education is shaped by usefulness, social influence, habit, facilitating conditions and local access constraints.
Present study	Qualitative interviews; Uganda; $N = 28$	Connects situated adoption to preferred PD and institutional conditions.

Note: N represents the number of participants in each category.

3. Methodology

3.1 Research Design

This study adopted an interpretive qualitative design to examine participants' experiences, meanings, and professional-development priorities. This design was appropriate because AI adoption is emerging, context dependent, and shaped by participants' roles, values, and institutional environments. The study was informed by a constructivist orientation: accounts were treated as situated interpretations rather than objective measurements of a single adoption reality. Reflexive thematic analysis was used because it supported systematic engagement with patterned meaning while recognizing the active role of researchers in interpretation (Braun & Clarke, 2022).

3.2 Study Area and Participants

The study was conducted in the Faculty of Education at a private university in Uganda. To protect the institution, it is referred to as University X. A convenience sample was recruited from lecturers and academic administrators who were available during the field visits and willing to discuss AI-related practices. Eligibility required current employment in the faculty and direct involvement in teaching, academic support, programme leadership, quality assurance, or research supervision. 28 participants were interviewed: 20 lecturers and 8 administrators. Variation was sought in gender, academic rank, years of service, and self-reported AI experience.

3.3 Interview Guide and Pilot Testing

A semi-structured interview guide was developed from the research questions and recent literature. It included questions on awareness and application of AI tools, perceived benefits, examples of teaching and research use, concerns, barriers, existing support, preferred professional-development formats, and recommendations for institutional policy. Probes elicited concrete examples and asked participants to distinguish between personal experimentation and institutionally supported use. The guide was pilot tested with two academic staff members from a different faculty who were not included in the final sample. The pilot indicated that participants interpreted “AI” as generative AI, so the guide was revised to define AI broadly while allowing detailed discussion of conversational systems. Questions were reordered to begin with practice before moving to policy and ethics.

3.4 Data Collection

Using the pilot-tested interview guide, trained interviewers met individually with 28 participants during 14 days of university visit in January, February, and March 2026. The dates of the fieldwork were 19–23 January, 16–20 February, and 9–12 March 2026. Interviews were conducted in private offices or meeting rooms and lasted 38–67 minutes. With consent, interviews were audio-recorded and supported by field notes. Participants were reminded not to disclose identifiable students’ information or confidential institutional data when describing AI use.

Data collection and preliminary analysis proceeded concurrently. The saturation judgement was qualitative rather than based on a fixed numerical threshold. As interviews accumulated, later accounts were compared with the developing codes and candidate themes to assess whether they introduced a substantively new dimension of meaning or elaborated patterns already present. Later interviews were repeating the established thematic structure without adding a substantively new dimension, and the research team judged that meaning saturation had been reached at the 25th interview. Three additional interviews were completed to test the adequacy and variation of the framework across participant roles and levels of AI experience. This approach is consistent with evidence that saturation depends on study scope, sample specificity, and analytic depth rather than a universal numerical threshold (Hennink & Kaiser, 2022).

3.5 Data Analysis

Audio recordings were transcribed verbatim, de-identified, and assigned participant codes P01–P28. Analysis followed the 6 recursive phases of reflexive thematic analysis: familiarization, initial coding, generating candidate themes, reviewing themes, defining and naming themes, and producing the report (Braun & Clarke, 2022). 2 researchers independently coded 6 transcripts to surface different interpretations and refine the coding framework; this process was used for analytic dialogue and reflexive discussion rather than for calculation of inter-rater reliability. The remaining transcripts were coded with continuing memo writing and comparison across role, experience, and gender. Candidate themes were checked against the full dataset, deviant cases were discussed, and the thematic map was revised until each theme had a coherent central organizing concept.

For descriptive transparency, participant-level thematic prevalence counts were generated after the thematic analysis had been completed. These counts were not treated as measures of statistical prevalence or as indicators of the relative importance of themes. Instead, they were used to show how widely a theme was represented across the 28 participant accounts. A participant was counted once only for each theme when their interview contained explicit and substantively relevant material contributing to that theme. Repeated references to the same theme within an individual interview did not increase the count. Participants could contribute to more than one theme; therefore, the theme frequencies were not mutually exclusive and were not expected to sum to 28.

To generate these descriptive counts, the researchers reviewed the coded material at participant level and recorded the presence or absence of each final theme in a participant-by-theme matrix. A value of 1 was assigned when the participant’s account contained sufficient substantive evidence to support inclusion within the theme, and a value of 0 was assigned when the theme was absent or when a reference was too incidental or ambiguous to constitute meaningful thematic contribution. Ambiguous cases were discussed during the analytic review and were counted only when the available interview material provided sufficient contextual evidence for inclusion. The

participant-level counts were checked against the relevant coded excerpts and interview accounts to ensure that each participant was counted no more than once within a given theme.

This procedure was applied consistently across all 6 themes: pragmatic but uneven adoption; AI as partner for efficiency and creativity; uncertainty about accuracy, authorship, and academic integrity; infrastructural and institutional constraints; demand for practice-based and discipline-relevant professional development; and the need for governance, communities of practice, and protected learning time. Thus, values such as 22/28, 26/28, 21/28, and 23/28 indicate the number of participants whose accounts contributed substantively to the respective themes, rather than the number of quotations, coded segments, or repeated mentions. The counts are therefore presented as illustrative prevalence within the qualitative sample and are not interpreted as statistical estimates or generalizable population proportions.

For interpretive transparency, AI literacy was not treated as a single binary code or equated with frequency of tool use. Instead, the analysis used the term as an analytic umbrella for participants' accounts of foundational understanding, prompting and task formulation, output verification, recognition of limitations and bias, privacy and authorship awareness, pedagogical judgement, and willingness to continue learning. These dimensions were interpreted in relation to, rather than mechanically imposed from, UNESCO's competency framework (UNESCO, 2024).

3.6 Trustworthiness and Reflexivity

Credibility was strengthened through prolonged engagement across three months, iterative probing, use of field notes, analytic meetings, and comparison of lecturers' and administrators' accounts. Dependability was supported by an audit trail containing versions of the guide, coding notes, theme definitions, and decision memos. Transferability was addressed through contextual description and a demographic profile. The researchers maintained reflexive notes about their own enthusiasm or concern regarding AI and considered how professional roles might influence participants' responses. A concise summary of the preliminary themes was shared with 4 participants for comment; their feedback confirmed the recognizability of the findings and added emphasis to the need for protected learning time.

3.7 Ethical Considerations

Participants were informed that participation was voluntary and that they could decline to answer any question or withdraw from the study before data anonymization. All identifying information was removed, and digital files were stored on encrypted, access-controlled devices. Quotations were edited minimally for readability without altering their intended meanings.

4. Results

4.1 Characteristics of Participants

The sample comprised 28 participants, including 20 lecturers and 8 administrators. 13 participants were identified as women and 15 as men; their ages ranged from 28 to 61 years. 6 participants held doctorates, 18 held master's degrees, and 4 held bachelor's degrees. 12 participants had fewer than 10-year of university experience, while 16 had 10 years or more. AI experience ranged from no direct use to daily use across multiple academic tasks. Table 2 presents the demographic profiles of the participants.

4.2 Theme 1: Pragmatic, Uneven Adoption

Participants described AI adoption as rapid, informal, and largely self-directed. 24 of the 28 participants had tried at least one generative AI tool. Of these, 9 described themselves as regular users and 4 as advanced or experimental users, giving 13/28 participants who reported regular or more advanced engagement with AI. Adoption was concentrated on low-risk or easily reversible tasks: generating lesson outlines, simplifying explanations, editing language, preparing examples, brainstorming research topics, producing draft emails, and creating quiz items. Few participants reported using AI for grading or consequential decisions, and those who did described it as advisory rather than determinative. "I started with it because I had three lectures to prepare for in one week. It gave me a structure, but I still had to bring in the Ugandan examples and correct many things" (P07, lecturer). "People are already using these tools, but mostly quietly. There is no common language for saying what is acceptable and what is not" (P22, administrator). Role-based comparison illustrated a difference in emphasis: lecturers described adoption more often through immediate teaching preparation and classroom tasks, whereas administrators more often framed the same informal uptake as a problem of institutional coordination, particularly the absence of shared expectations about acceptable use.

Experience differed by discipline, age, digital confidence, and access to colleagues who were already experimenting. Some participants framed AI as an extension of Internet search or word processing; others viewed it as qualitatively different because it could produce coherent academic text. 4 participants had not used AI directly, citing lack of confidence, limited need, or concern about becoming dependent on a system they did not understand.

Table 2. Demographic characteristics of participants

Characteristic	Category	N	%
Gender	Women	13	46.4
	Men	15	53.6
Role	Lecturer	20	71.4
	Administrator	8	28.6
Age	20–29	3	10.7
	30–39	8	28.6
	40–49	9	32.1
	50+	8	28.6
Highest qualification	Bachelor/Professional	4	14.3
	Master	18	64.3
	Doctorate	6	21.4
	<5 years	6	21.4
University experience	5–9 years	6	21.4
	10–14 years	7	25.0
	15+ years	9	32.1
Self-rated artificial intelligence (AI) use	Non-user/Observer	4	14.3
	Occasional	11	39.3
	Regular	9	32.1
	Advanced/Experimental	4	14.3

Note: *N* represents the number of participants in each category.

4.3 Theme 2: Artificial Intelligence as an Efficiency and Creativity Partner

Participants valued AI primarily for saving time and helping them begin difficult tasks. 22 of the 28 participants explicitly discussed timesaving or efficiency-related uses of AI, including assistance with lesson preparation, drafting, summarization, generation of examples, and routine academic tasks. They described the “blank page” as a major barrier in course preparation and academic writing. AI-generated drafts, outlines, examples, and alternative explanations created a starting point that users could revise. Participants also used AI to adapt text for different student levels, draft case scenarios, and generate questions for classroom discussion. “It does not replace my thinking. It gets me moving. I ask for 5 examples, reject three, change one, and the last one may trigger a better idea” (P14, lecturer). Research-related uses included refining keywords, explaining statistical concepts, summarizing pasted non-confidential text, generating interview probes, and improving grammar. Administrators reported using AI for meeting agendas, routine correspondence, concept notes, and policy comparison. Nonetheless, participants repeatedly stressed that efficiency depended on verification and domain expertise; inexperienced users could accept fluent but inaccurate output. The efficiency benefits also reflected participants’ roles. Lecturers most often linked time saving and creativity to lesson preparation, examples or assessment tasks, and research support, while administrators would rather describe gains in routine correspondence, meeting preparation, concept notes, and policy comparison.

4.4 Theme 3: Accuracy, Authorship, and Academic Integrity Uncertainty

Concerns about accuracy, authorship, and academic integrity were widespread. 26 of the 28 participants raised at least one substantive concern about reliability, authorship, academic integrity, or related risks associated with AI use. The most pervasive concern was uncertainty about what could be trusted and what should be disclosed. Participants had encountered fabricated references, incorrect definitions, and confident but contextually inappropriate answers. This produced a paradox: AI could save time, but checking outputs sometimes consumed the time saved. Lecturers also worried that students might submit AI-generated assignments without understanding them. “The language is convincing, and that is the danger. A student can bring a very polished answer, but when you ask one follow-up question, there is no ownership of the ideas” (P03, lecturer). “We need rules that distinguish assistance from substitution. Editing grammar is not the same as asking the system to write the whole dissertation chapter” (P25, administrator). Participants did not support a simple ban. They believed prohibition would drive use underground and leave students without guidance. Instead, they wanted clear disclosure requirements, examples of acceptable and unacceptable use, and assessment approaches that made learning processes visible. Suggested strategies included oral defense, staged submissions, reflective commentaries, version histories,

localized tasks, and in-class application. Although both groups raised integrity concerns, lecturers more often focused on student learning, assessment ownership, and the difficulty of judging whether submitted work reflected understanding. Administrators usually emphasized the need for institution-wide categories of acceptable use, disclosure expectations, and consistent rules across programmes.

4.5 Theme 4: Infrastructural and Institutional Constraints

AI adoption was shaped by familiar digital constraints but also by recent cost and governance problems. 21 of the 28 participants discussed at least one infrastructural or institutional constraint affecting AI adoption, including internet connectivity, cost, device access, electricity reliability, subscription fees, data governance, or institutional policy. Participants mentioned unstable internet connectivity, the price of mobile data, limited access to capable devices, electricity interruptions, and subscription fees for advanced models. Free versions were seen as useful but inconsistent, while paid services created unequal capability among staff and students. “We cannot make a policy assuming everyone has the premium version and unlimited data. That would create another digital divide inside the same classroom” (P11, lecturer). Institutional uncertainty was equally important. Participants were unsure whether they were permitted to upload draft teaching materials, student work, or research data to external platforms. They also lacked guidance on copyright, data retention, and procurement. Several administrators noted that decisions were being postponed because tools were changing quickly, yet participants argued that waiting for stability was itself a risky policy because of current use. Here again, the groups converged on the existence of constraints but differed in emphasis: lecturers tended to describe the practical consequences of data costs, connectivity, devices, and unequal access for teaching and student participation, whereas administrators usually foregrounded governance uncertainty, procurement, handling of data, and the difficulty of setting policy while tools were changing rapidly.

4.6 Theme 5: Practice-based, Differentiated Professional Development

All 28 participants expressed a need for professional development or training related to AI use, although they differed in the level, content, and preferred delivery format. Participants rejected generic awareness seminars owing to their insufficiency. They wanted hands-on sessions linked to authentic academic tasks, with opportunities to compare weak and strong outputs, practise verification, and discuss disciplinary cases. Beginners requested foundational orientation, while experienced users wanted assessment redesign, research workflows, automation, and ethical case analysis. “Do not only show us 10 tools in 2 hours. Let me bring my course outline, improve one activity, and leave with something I can use on Monday” (P18, lecturer). “Training should have levels. Some colleagues need to know what a prompt is; others are ready to build a departmental workflow. Putting everyone in one workshop frustrates both groups” (P27, administrator). Preferred formats included short modular workshops, peer demonstrations, mentoring, drop-in clinics, recorded low-bandwidth tutorials, and communities of practice. Participants wanted examples from education, supervision, assessment, and research rather than generic business examples. They also requested guidance on local relevance, inclusive practice, and how to teach future teachers to use AI responsibly. Lecturers generally described this need in terms of authentic teaching, supervision, assessment, and research tasks. Administrators shared the preference for practical training but more frequently stressed differentiated pathways, institutional consistency, and the need to connect staff development with policy and support structures.

4.7 Theme 6: Governance, Communities of Practice, and Protected Time

Participants understood professional development as an organizational responsibility rather than an individual hobby. 23 of the 28 participants explicitly requested ongoing institutional support, including access to approved tools, designated support people, communities of practice, clear guidance, or protected time for professional development. Without protected time, even motivated staff prioritized teaching, marking, supervision, and administration over experimentation. “The problem is not willingness. The problem is that learning AI is added on top of everything else. Give us two protected afternoons and someone to support us, and the uptake will be different” (P05, lecturer). Participants proposed a Faculty AI community of practice that would meet monthly, share tested prompts and activities, document failures, and advise management. Administrators saw such a group as a bridge between policy and practice. They also emphasized that policy should be reviewed regularly and included student representation because classroom norms depend on mutual transparency. The role comparison was complementary rather than oppositional: lecturers emphasized protected time and accessible support as conditions for participation, while administrators positioned communities of practice as a bridge between emerging policy and day-to-day academic practice.

6 interrelated themes emerged from the interviews. The participant-level thematic review indicated that 24 of 28 participants had tried at least one generative AI tool, with 13 reporting regular or advanced engagement. 22

participants discussed time-saving or efficiency-related uses, while 26 raised concerns about accuracy, authorship, or academic integrity. 21 participants identified infrastructural or institutional constraints, and all 28 participants expressed a need for professional development or training. In addition, 23 participants requested ongoing institutional support, including communities of practice, approved tools, designated support, clear governance, or protected learning time. These counts are descriptive indicators of the breadth of representation of the themes within the qualitative sample and are not statistical estimates. Because participants could contribute to multiple themes, the frequencies are not mutually exclusive. Table 3 summarizes the 6 themes, their illustrative participant-level prevalence, and their professional-development implications.

Table 3. Summary of emergent themes, illustrative prevalence, and professional-development implications

Theme	Illustrative Prevalence	Central Professional-Development Implication
Pragmatic but uneven adoption	24/28 tried artificial intelligence (AI); 13/28 reported regular or advanced engagement	Differentiate entry-level and advanced pathways.
Efficiency and creativity partner	22/28 discussed timesaving or efficiency-related uses	Teach task selection, iteration, and human review.
Accuracy, authorship, and academic integrity	26/28 raised reliability, authorship, or integrity concerns	Prioritize verification, disclosure, and assessment redesign.
Infrastructure and institutional constraints	21/28 discussed access, cost, connectivity, data, or policy barriers	Provide approved tools, low-bandwidth options, and clear governance.
Practice-based professional development	28/28 expressed a need for AI-related training or professional development	Use authentic tasks, discipline examples, coaching, and modular delivery.
Governance, communities of practice, and protected time	23/28 requested ongoing institutional support	Create communities of practice and recognize learning in workload.

Note: Illustrative prevalence refers to the number of participants whose interview accounts contributed explicitly to each theme. Because participants could contribute to more than one theme, frequencies are not mutually exclusive and should not be interpreted as statistical estimates.

5. Discussion

5.1 Use-Driven but Institutionally Underdeveloped Adoption

Within the Faculty studied, the findings portrayed AI adoption as a bottom-up process driven by immediate work demands. Lecturers adopted AI where benefits were visible and risks seemed manageable, especially for planning, language support, and idea generation. This aligns with international evidence that faculty adoption varies by confidence, perceived usefulness, and task, rather than occurring as a uniform institutional shift (Mah & Groß, 2024). In this particular private Ugandan university setting, the fragility of such adoption was visible in participants' reliance on personal data, individual devices, free accounts, and informal peer support. These patterns should therefore be read as context-specific evidence rather than as a description of Ugandan higher education as a whole.

This pattern suggests that adoption should not be measured only by whether lecturers have used a tool. Responsible adoption includes the ability to select appropriate tasks, evaluate output, explain decisions, protect data, and align use with learning outcomes. A lecturer who frequently uses AI without verification may be less professionally prepared than a cautious user who understands its limitations. Professional development should therefore target judgement and practice, not only frequency of use. The comparison between lecturers and administrators also indicates that responsible adoption is role-sensitive. Lecturers encountered AI primarily at the point of teaching, assessment, supervision, and research practice, while administrators more often encountered it through policy, coordination, and institutional risk. Professional development should therefore differentiate not only by experience level but also by role and responsibility.

5.2 Efficiency Requires Epistemic Vigilance

Participants' accounts of efficiency were consistently qualified by the need for checking. This supports literature describing generative AI as simultaneously productive and unreliable (Kasneci et al., 2023; Lo, 2023). Fluency could conceal error, and this is especially consequential in teacher education because inaccurate content may be reproduced in schools. The professional competence at stake is epistemic vigilance: knowing when AI is likely to be useful, what evidence is required, and how to triangulate claims with authoritative sources.

Training should make hallucination and bias observable through exercises rather than simply warning participants. For example, lecturers could compare AI-generated references with database records, test the same prompt across systems, identify missing local perspectives, and revise outputs using disciplinary criteria. Such

activities transform abstract caution into a repeatable verification routine.

5.3 Academic Integrity Addressed Through Learning Design

Participants' rejection of blanket bans reflects a wider shift from policing toward redesign. Generative AI complicates conventional take-home assignments that reward a polished product without evidence of process. Professional development should help lecturers clarify the purpose of each assessment and decide where AI uses support or undermines that purpose. Transparent categories, i.e., prohibited, permitted with disclosure, encouraged or required, could be tied to learning outcomes and communicated in course outlines.

Assessment redesign need not eliminate written work. It could combine staged drafts, annotated sources, oral explanation, local data, reflective accounts of AI use, and demonstration of decision making. These approaches preserve human judgement and reduce the value of undisclosed substitution. They also align with UNESCO's emphasis on human agency and responsible use (UNESCO, 2023).

5.4 A Contextualized Model of Artificial Intelligence Professional Development

The 6 themes supported a 7-component professional-development model for the faculty: (1) foundational AI literacy; (2) prompt design and verification; (3) pedagogical integration; (4) assessment and academic integrity; (5) research, authorship, and data protection; (6) equity, bias, and local relevance; and (7) ongoing peer learning and technical support. In this model, foundational AI literacy does not mean prompt proficiency alone. It includes a basic understanding of AI capabilities and limitations, the ability to frame appropriate tasks, critical verification of outputs, recognition of bias and uncertainty, awareness of privacy and authorship implications, and informed decisions about when AI use is pedagogically appropriate. These elements correspond broadly to UNESCO's competency dimensions but are tailored to the immediate practices and constraints reported by participants (UNESCO, 2024).

Delivery should be modular and differentiated. A foundation pathway can introduce concepts, approved tools, basic prompting, verification, and privacy. An applied teaching pathway could focus on learning outcomes, lesson activities, feedback, and assessment. A research pathway could address literature discovery, analysis support, authorship, confidential data, and citation verification. An advanced pathway could support workflow design, discipline-specific experimentation, and faculty leadership. Each pathway should culminate in a practical artefact reviewed by peers.

Sustainability requires institutional conditions: protected time, leadership endorsement, access to appropriate tools, help mechanism, and periodic policy review. Communities of practice could capture local knowledge more effectively than isolated workshops because AI tools and norms evolve rapidly. They could also help relieve anxiety by normalizing discussion of failures and uncertainty.

5.5 Implications for Policy and Leadership

For the institution studied, the findings support development of a concise AI framework through consultation with academic staff, students, information technology personnel, librarians, legal or data-protection officers, and research ethics structures. Such a framework could specify data categories that must not be entered into public systems, expectations for disclosure, responsibility for verification, acceptable assessment uses, procurement criteria, and procedures for updating guidance. More broadly, these recommendations may be useful for comparable institutions, but their applicability should be assessed against local governance arrangements, resources, disciplines, and student populations. Equitable adoption may also require institutional licenses or alternatives that do not privilege those able to pay.

For the Faculty of Education studied, participants' accounts suggest that AI competence should be integrated into teacher preparation and lecturer development. Lecturers need support not only to use AI themselves but to model responsible use for future teachers. This includes discussing bias, language and cultural relevance, accessibility, environmental cost, and the social consequences of automated decision making. Whether the same priorities hold in other Ugandan faculties or universities is an empirical question, particularly in disciplines with different assessment cultures, technical infrastructures, and professional accreditation requirements.

5.6 Theoretical Contribution

The study contributes a situated account of AI adoption as professional practice within one Faculty of Education. It suggests that adoption is not necessarily a linear movement from non-use to use but can involve a negotiated configuration of task value, confidence, verification burden, infrastructure, institutional legitimacy, and ethical risk. In this setting, professional development functioned in participants' accounts as a potential mediating structure between individual experimentation and responsible organizational capability. The findings also suggest

that, in resource-constrained settings, infrastructure and governance may shape what AI adoption means and who can participate rather than operating only as external barriers. These propositions require testing and refinement across other institutions, sectors, and disciplines.

6. Conclusions

This study explored AI adoption and professional-development needs among 28 lecturers and administrators in the Faculty of Education at one private Ugandan university. Participants described growing but uneven use of AI for teaching preparation, academic writing, research support, communication, and administration. They valued speed, idea generation, and language support but remained concerned about inaccurate output, fabricated sources, authorship, student overreliance, privacy, unequal access, and the absence of clear institutional guidance. Role-based comparison added nuance: lecturers more often foregrounded classroom, assessment, supervision, and workload concerns, whereas administrators usually emphasized policy gaps, governance consistency, procurement, and institutional support.

For this Faculty, the findings indicated that professional development should be practical, differentiated, discipline relevant, role-sensitive, and sustained. One-off awareness sessions are unlikely to produce responsible practice. Lecturers and administrators need opportunities appropriate to their responsibilities: authentic teaching and research tasks, verification routines, assessment redesign, discussion of ethical case, policy interpretation, and governance planning. The institution should complement training with approved tools, governance, technical support, communities of practice, and protected learning time.

The central conclusion from this setting is that responsible AI adoption is a professional and organizational learning challenge, not merely a matter of tool access. Faculties of Education may have a strategic role because lecturer practices influence both university learning and the preparation of future teachers. A human-centred approach that combines AI literacy, pedagogical judgement, research integrity, equity, and institutional support offers a promising basis for sustainable adoption. However, these conclusions are context-specific and should not be interpreted as representatives of Ugandan higher education on the whole. Patterns may differ in public universities, other private institutions, and faculties such as engineering, medicine, or the natural sciences, where infrastructure, disciplinary practices, governance arrangements, and assessment cultures may differ.

7. Limitations and Future Research

The study is limited by its single-faculty, single-institution setting, and convenience sampling. Participants who volunteered may have been more interested in AI than non-participants. Self-reported practice may also differ from observable use. The findings are therefore analytically transferable rather than statistically generalizable: they illuminated patterns within this private Ugandan Faculty of Education but could not establish that the same patterns characterize public universities, other private institutions, or disciplines with different technical and pedagogical cultures. Because AI tools and policies change rapidly, the findings also represented a time-bound account. Future research should include multiple Ugandan universities, compare public and private institutions, examine disciplinary differences, and evaluate professional-development interventions longitudinally. Studies should also investigate students' perspectives, local-language use, costs, data governance, and the relationship between AI-supported teaching practices and learning outcomes.

Author Contributions

Conceptualization, A.U.; methodology, A.U. and N.J.; investigation, A.U. and N.J.; writing—original draft preparation, A.U. and N.J.; writing—review and editing, A.U. and N.J.; visualization, A.U. and N.J. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

All participants provided their written consent to take part in this study.

Ethical Approval

Kampala International University Research Ethics Committee granted the study ethical approval under reference number: KIU-2026-036.

Data Availability

The datasets generated during and/or analysed during the current study are available from the corresponding

author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

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